

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Fundamentals of Natural Language Processing	
Course code	CS3233	
Semester:	VI	
Names of the Course Instructor(s)	Dr. Narendra, Prof. Khushboo Pandey, Dr. Saliha Bathool, Prof. Priyanka S	
Course credit	No. of hours per week (2+0+2)	Total no. of Teaching hours
3	04 Hours	60 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	Introduction to NLP: tasks, pipeline, applications	Interactive PPT with real-world examples; think-pair-share on daily NLP usage	Jurafsky & Martin Ch.1
2.		Text preprocessing: tokenization, normalization, sentence segmentation	Live Python demo using spaCy/NLTK	Jurafsky & Martin Ch.2; Bird et al. Ch.3
3.		Lemmatization, POS tagging and NER: concepts and tagsets	Chalk-and-talk + code walkthrough, Interactive PPT with tagging exercises	Eisenstein Ch.2, Jurafsky & Martin Ch.10
4.		Language modelling: n-grams, probability, perplexity	Problem-solving on board; mini-quiz	Jurafsky & Martin Ch.3
5.		Traditional Representations: BoW, TF-IDF and limitations	Comparison activity: BoW vs TF-IDF on sample corpus	Eisenstein Ch.4; Rao & McMahan Ch.2
6.		Contextual embeddings: ELMo, BERT embeddings (conceptual)	Concept PPT + visualizations	Tunstall et al. Ch.2
7.		Introduction to NLP Lab	Overview of NLP lab objectives, datasets, NLP	Python, Jupyter, NLTK, spaCy

			pipeline, environment setup, and Python NLP ecosystem	
8.		Text Tokenization	Sentence and word tokenization; comparison of rule-based and statistical tokenizers	NLTK, spaCy
9.		Stop-word Removal & Lemmatization	Stop-word filtering, stemming vs lemmatization, vocabulary reduction analysis	spaCy, NLTK
10.		Part-of-Speech Tagging	POS tagging concepts, tag sets, tagging accuracy analysis	spaCy, NLTK
11.		Named Entity Recognition	Identification and visualization of named entities; entity-level error analysis	spaCy
12.		Bag of Words (BoW)	Document-term matrix construction and sparsity analysis	scikit-learn
13.		Modern embedding applications & challenges (bias, multilingual)	Case-study discussion	Jurafsky & Martin bias section
14.	2	Neural networks recap; CNNs for text	PPT + whiteboard derivations	Goodfellow et al. Ch.6; Goldberg Ch.5
15.		RNNs and vanishing gradients	Derivation + sequence examples	Goldberg Ch.6
16.		LSTM and GRU architectures	Step-by-step gate illustration	Goldberg Ch.7
17.		Attention mechanism for sequences	Visual explanation; small math example	Vaswani et al. summary; Tunstall et al. Ch.3
18.		Evaluation metrics: precision, recall, F1, confusion matrix	Problem-solving session	Eisenstein Ch.8
19.		Evaluation metrics: BLEU, ROUGE, Use Case: Text classification (news, emails, etc.)	Worked examples for MT and summarization	Koehn Ch.7
20.		TF-IDF Representation	Computation of TF-IDF vectors and feature importance analysis	scikit-learn
21.		Distributed Word Representations	Introduction to Word2Vec and GloVe; loading pretrained	gensim

			embeddings	
22.		Word Representation Comparison	Comparative analysis of BoW, TF-IDF, and embeddings using similarity tasks	scikit-learn, gensim
23.		Introduction to Sequence Models	Sequential nature of text data; dataset exploration for sentiment analysis	Python, Keras / PyTorch
24.		LSTM-based Text Classification	Design and implementation of LSTM-based sentiment classifier	Keras / PyTorch
25.		GRU-based Text Classification	Implementation and comparison of GRU-based classifier	Keras / PyTorch
26.		Motivation for transformers; seq2seq limitations	Concept PPT; discussion	Tunstall et al. Ch.3
27.		Transformer encoder-decoder architecture (multi-head attention, positional encoding)	Diagram-based explanation	Vaswani et al.; Tunstall Ch.4
28.		Pretrained encoders: BERT, RoBERTa, ALBERT	Comparison table + use-cases	Tunstall Ch.5
29.		Generative models: GPT-2/3, T5; prompt-based learning	Demo and discussion on capabilities and risks	Tunstall Ch.8
30.	3	Transfer learning and fine-tuning strategies; overfitting, domain shift	Chalk-and-talk + case-study	Tunstall Ch.6
31.		HuggingFace Transformers Library.	Short lecture + Coding	Recent articles on green AI
32.		Model Evaluation & Optimization	Performance evaluation, hyperparameter tuning, and error analysis	scikit-learn
33.		CNN for Text Classification	Convolutional architectures for text; n-gram feature learning	TensorFlow / PyTorch
34.		Attention Mechanism	Implementation of attention layer and visualization of attention weights	Keras / PyTorch

35.		Model Comparison Study	Comparative study of CNN, RNN, and attention-based models	TensorFlow / PyTorch
36.		Introduction to Transformers	Transformer architecture, self-attention, and pretrained models	Hugging Face Transformers
37.		Fine-tuning BERT for Text Classification	Dataset preparation and fine-tuning BERT on classification tasks	Transformers, Datasets
38.		Question Answering (QA) Systems: Closed-domain QA and Open-domain QA	PPT + live QA demo	Jurafsky & Martin QA section
39.		Conversational AI: Dialogflow (by Google), Open-source Conversational Framework (Rasa)	Whiteboard design of sample chatbot	Rasa docs overview
40.		Summarization: Extractive and Abstractive	PPT + analysis of model outputs	Koehn Ch.10
41.		Machine Translation: OpenNMT, MarianMT	Concept PPT	Koehn Ch.1–4
42.	4	Real-World Case Studies: Healthcare, Finance, Legal NLP	Group presentations	Selected case-study papers
43.		BERT for Named Entity Recognition	Token classification using BERT and subword alignment	Transformers
44.		Language Generation	Text generation and paraphrasing using GPT-2 or T5	Transformers, Pipeline
45.		Question Answering System	Implementation of extractive QA using pretrained BERT models	Transformers, pipeline
46.		Text Summarization	Abstractive summarization using BART or T5	Transformers
47.		Machine	Multilingual translation using	Transformers

		Translation	MarianMT or mBART	
48.		Speech-to-Text	Automatic speech recognition using Whisper or Wav2Vec 2.0	Whisper, Transformers
49.		Automatic Speech Recognition (ASR) using Wav2Vec 2.0 (by Facebook AI), Whisper (by OpenAI).	PPT; audio demos	Jurafsky & Martin speech chapters
50.		Automatic Speech Recognition (ASR) using Whisper (by OpenAI).	PPT; audio demos	Jurafsky & Martin speech chapters
51.		TTS fundamentals; Tacotron2 and FastSpeech2 (concepts)	Diagram-based explanation	Recent TTS tutorials
52.		Multimodal NLP: CLIP, LLaVA, applications	PPT + demo	Recent CLIP/LLaVA docs
53.	5	NLP for Low-Resource Languages Overview - Multilingual BERT (mBERT)	Case-study discussion	Multilingual NLP articles
54.		Cross-lingual RoBERTa (XLM-R), IndicNLP	Case-study discussion	Multilingual NLP articles
55.		Text-to-Speech	Text-to-speech synthesis using Tacotron or FastSpeech	Coqui TTS, PyTorch
56.		Multimodal NLP	Image-text understanding using CLIP or LLaVA	OpenCLIP, LLaVA
57.		Mini Project – Proposal	Problem formulation, dataset selection, and baseline planning	Any relevant tools
58.		Mini Project – Model Development	Model training, experimentation, and optimization	Student choice
59.		Mini Project – Evaluation	Performance evaluation, result analysis, and report drafting	scikit-learn

60.	Mini Project – Final Demo & Viva	Project demonstration, viva voce, and final report submission	—
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Assessment and Evaluation

Continuous Internal Evaluation (CIE): 70%	
Components of CIE	Weightage of each component
CIE-1 - Theory Quiz Component 1(Module 1 and the first half of Module 2) - Laboratory Assessment (Programs 1–4) - Literature Survey + Plagiarism Report	5 marks out of 70 5 marks out of 70 10 marks out of 70
CIE-2 (Coverage: Modules 1 to 3) - Theory Pen-and-paper	25 marks out of 70
CIE-3 - Laboratory Assessment (Programs 5–8) - Laboratory Assessment (Programs 9–12) - Mooc Assignment - Project Manuscript (Conference Paper) and Demonstration	5 marks out of 70 5 marks out of 70 5 marks out of 70 10 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of Exam	30%

Course Outcomes- Programme Outcomes Matrix

(3 = highly related, 2 = moderately related, 1 = slightly related, - = not related)

Sl.	Course	P	P	P	PO	PO	PO	PO	PO	PO	PO	PO	PO
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No	outcome	O1	O2	O3	4	5	6	7	8	9	10	11	12
1.	Apply foundational and contextual NLP techniques using linguistic preprocessing, word embeddings, and classical models.	3	3	1	2	1	1	1	1	1	1	1	1
2.	Design and implement deep learning architectures for sequence modelling and text classification tasks.	3	3	2	3	1	1	1	2	1	1	1	2
3.	Analyze, fine-tune, and deploy transformer-based models for advanced NLP tasks such as sentiment analysis, question answering, and summarization.	3	3	3	3	2	1	2	3	2	3	2	2
4.	Develop and integrate real-world NLP applications including speech-based systems, machine translation, dialogue systems, and socially relevant multilingual solutions.	3	2	3	3	3	3	3	3	3	3	3	2

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1 (5 Marks)	Theory Quiz (Module 1 and the first half of Module 2) will be conducted for 5 marks. Each right answer carries one mark. No negative marking for wrong answers.
	Component 2 (5 Marks)	Laboratory Assessment (Programs 1–4) (<i>Refer Table-1</i>)
	Component-3 (10 Marks)	Literature Survey + Plagiarism Report

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory (25 Marks)	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component 1 (5 Marks)	Laboratory Assessment (Programs 5–8) = 5 Marks (<i>Refer Table-1</i>)
	Component 2 (5 Marks)	Laboratory Assessment (Programs 9–12) = 5 Marks (<i>Refer Table-1</i>)
	Component 3 (5 Marks)	Mooc Assignment = 5 Marks
	Component 4 (10 Marks)	Project Manuscript (Conference Paper) + Demonstration = 10 Marks (<i>Refer Table-2</i>)

Table-1

Lab Write-up and Execution (Max: 5 marks)
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Sl. No	Criteria	Measuring Methods	Excellent (1 Mark)	Good (0.5 Mark)	Poor (0 Mark)
1.	Understanding of problem statement. (1 Marks)	Observations	Student exhibits thorough understanding of requirements and applies suitable algorithm for the problem. (2 M)	Student has sufficient understanding of requirements and applies suitable algorithm for the problem. (<2 M and >=1 M)	Student does not have a clear understanding of requirements and is unable to apply suitable algorithm for the problem. (0 M)
2.	Execution (2 Marks)	Observations	Student demonstrates the execution of the program with optimized code and shows performance efficiency. Appropriate validations with all test cases are handled. (2 M)	Student demonstrates the execution of the program without optimization of the code and shows performance efficiency with only few test cases. (1 M)	Student has not executed the program. (0 M)
3.	Results and Documentation on GCR (2 Marks)	Observations	Documentation with appropriate comments and all output screenshots is shown. (2 M)	Documentation with only few comments and only few output screenshots is shown. (1 M)	Documentation with no comments and no output screenshots is shown. (0 M)

Table-2:

**CIE 3 Graded Component – 3 Project Manuscript (Conference Paper) + Demonstration
(15 Marks)**

Assess students' performance through continuous Project work and their understanding via presentation.

SN.	Criteria	Description	Excellent	Satisfactory	Needs Improvement	Max Marks
1	Problem definition & literature review	Clarity of problem statement, motivation, and coverage of related work	Problem is very clearly defined and well motivated; literature review is recent, relevant, and critically analyzed; clear research gap identified (2 Marks)	Problem reasonably defined; literature review covers some relevant work but with limited critique or missing a clear gap (1 Mark)	Problem unclear or trivial; literature review very shallow or mostly missing; no gap identified(0 Mark)	2
2	Methodology & technical correctness	Soundness of approach, model design, experiments, and use of NLP tools	Methodology is appropriate and well justified; model/algorithm s correctly implemented; experimental design is systematic and reproducible (3 Marks)	Methodology generally appropriate but with minor gaps; implementation mostly correct; experiments limited in scope or not fully systematic (2 Marks)	Methodology inappropriate or poorly described; major implementation errors; experiments missing or invalid(0 Mark)	3

3	Results, evaluation & discussion	Quality of results, metrics, comparison and interpretation	Meaningful quantitative results with appropriate metrics; comparisons/baselines provided; insightful discussion of strengths, limitations, and error analysis (3 Marks)	Some quantitative results with basic metrics; limited comparison; discussion mostly descriptive with little analysis (2 Marks)	Very few or no results; metrics inappropriate; no meaningful discussion or analysis (0 Mark)	3
4	Manuscript structure, writing & formatting	Organization, clarity, figures/tables, and adherence to conference template	Manuscript follows conference template closely; sections logically organized; language clear and concise; figures/tables well labelled and referenced (2 Marks)	Manuscript mostly follows template with minor deviations; organization acceptable; language understandable but with some issues; figures/tables basic (1 Mark)	Manuscript poorly structured; template not followed; many language issues; missing or unclear figures/tables (0 Mark)	2
5	Oral presentation & live demonstration	Clarity of presentation, time management, demo quality, and ability to answer questions	Presentation is clear, well-timed and engaging; demo runs smoothly and shows key features; questions answered confidently and	Presentation somewhat clear but rushed/over-time; demo works with minor issues; answers to questions partially correct (3 Marks)	Presentation unclear or disorganized; demo fails or is missing; unable to answer questions (1 Marks)	5

			accurately (5 Marks)			
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